SKILLBRIDGE PLATFORM

USER MANUAL

End-to-End Operational Guide for Students, Faculty Mentors, Recruiters, and Platform Administrators

Document Version:	1.0.0 (Production Release)
Software Release:	SkillBridge Academic Placement Suite
Author / Engineering Team:	SkillBridge Core Development Team
Target Audience:	Students, Faculty Guides, Placement Officers, Enterprise Recruiters
Publication Date:	November 2026
Document Classification:	Official Institutional Documentation
Primary Application URL:	http://localhost:5173

DOCUMENT PURPOSE: This user manual provides step-by-step guidance for navigating and operating all four dedicated role-based portals in SkillBridge: Student Workspace, Mentor & Faculty Workspace, Recruiter Hub, and the Administrator Console.

Table of Contents

1. System Overview & Platform Mission	**Page 3**
2. System Requirements & Browser Setup	**Page 3**
3. Accessing the Application & Network Endpoints	**Page 4**
4. Authentication, Registration & Session Management	**Page 4**
5. Role-Based Security & Workspace Compartmentalization	**Page 6**
6. Student Workspace (8 Step-by-Step Functional Modules)	**Page 7**
6.1 Student Overview & Command Center	**Page 7**
6.2 Student Profile & Academic Credentials	**Page 8**
6.3 Skill Evidence Portfolio & AI OCR Extraction	**Page 9**
6.4 My Skills Taxonomy Inventory	**Page 11**
6.5 Skill Gap Analysis & Deficit Detection	**Page 12**
6.6 Curated Learning Recommendations	**Page 13**
6.7 Job Role Benchmark Matching	**Page 14**
6.8 Overall Readiness & Longitudinal Progress	**Page 15**
7. Faculty & Mentor Workspace (Review Queue, Cohort & Analytics)	**Page 16**
8. Recruiter Hub (Job Requisitions, Candidate Discovery & Shortlists)	**Page 17**
9. Platform Administrator Console (Users, Health, Audit Trail)	**Page 18**
10. Troubleshooting, Error Codes & Frequently Asked Questions	**Page 19**
11. Technical Support & Verification Compliance Sign-off	**Page 20**

1. System Overview & Platform Mission

SkillBridge is an AI-powered student skill intelligence and placement acceleration platform designed for engineering and technical colleges. The platform bridges the gap between academic course curriculum and industry hiring standards by automatically parsing real student evidence (such as course certificates, coding profiles, proctored examinations, and Git repositories), mapping demonstrated skills against a standardized technical taxonomy, calculating empirical role-readiness scores, and providing personalized skill-gap remediation pathways.

SkillBridge enforces four distinct workspaces: **Student**, **Faculty Mentor**, **Enterprise Recruiter**, and **System Administrator**, ensuring that each user role receives specialized analytics, workflows, and strict data governance.

2. System Requirements & Browser Setup

Component	Minimum Specification	Recommended Specification
Web Browser	Google Chrome 100+, Edge 100+, Firefox 90+	Latest Google Chrome or Microsoft Edge
Screen Resolution	1280 x 720 (HD)	1920 x 1080 (Full HD) or higher
JavaScript	Enabled (ES2022+)	Enabled with Web Workers
Network Connection	Localhost (Offline) or 1 Mbps LAN	High-speed Campus Intranet

3. Accessing the Application & Network Endpoints

SkillBridge is accessed via any standard desktop web browser. When launched locally, the frontend client connects transparently to the backend API Gateway and AI microservice.

Service	Host & Port	Primary Function
Frontend Client	http://localhost:5173	User interface, dashboard views, forms, and charts
API Gateway	http://localhost:5000	Authentication, CRUD operations, database queries
AI Microservice	http://localhost:8000	Optical Character Recognition, NLP skill extraction

4. Authentication, Registration & Session Management

SkillBridge uses JSON Web Token (JWT) based authentication. Passwords are encrypted using salted bcrypt hashes. Upon entering valid credentials, users receive an access token that securely validates subsequent API calls.

4.1 Sign-In Procedure & Pre-configured Accounts

1. Open your browser and navigate to http://localhost:5173/login.
2. Enter your institutional email address and password, then click **Sign In**.
3. Alternatively, click any of the **Demo accounts** cards at the bottom of the form to automatically pre-fill credentials for immediate evaluation:

Role Persona	Email Address	Password	Target Workspace
Faculty Guide	faculty@skillbridge.edu	Faculty@123	Mentor Workspace (/faculty)
Platform Admin	admin@skillbridge.edu	Admin@123	Admin Console (/admin)
Demo Student	student@skillbridge.edu	Student@123	Student Workspace (/student/dashboard)
Recruiter	recruiter@skillbridge.edu	Recruiter@123	Recruiter Hub (/recruiter)

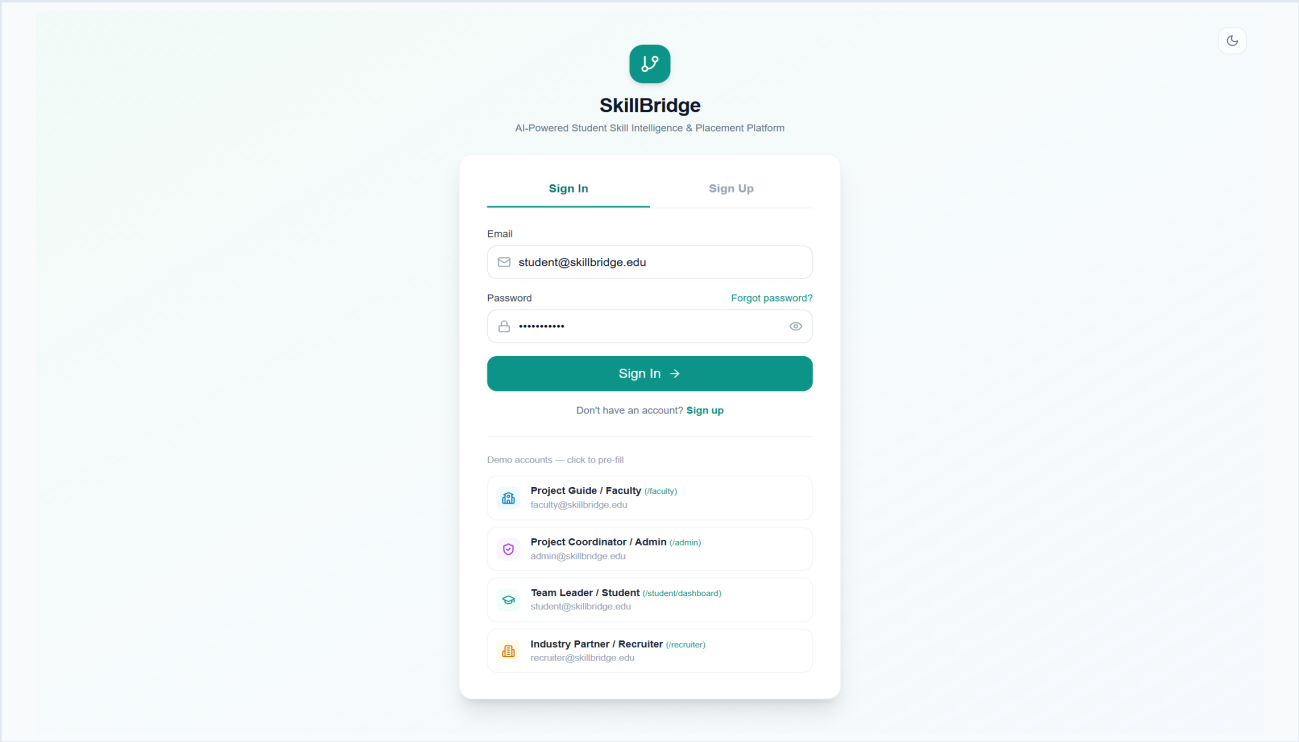


Figure 1: SkillBridge Unified Sign-In Screen with 1-Click Persona Pre-Fill Buttons

4.2 Student Account Self-Registration

New students can create their own account by clicking **Sign Up** on the login screen or navigating to /register:
1. Enter your full name (e.g. *Om Ingale*).
2. Provide your academic email address (e.g. *omingale26@gmail.com*).
3. Set a secure password with a minimum of 6 characters.
4. Click **Create Student Account**. The system initializes your student profile, creates a unique candidate alias, and navigates directly to your personalized Student Workspace.

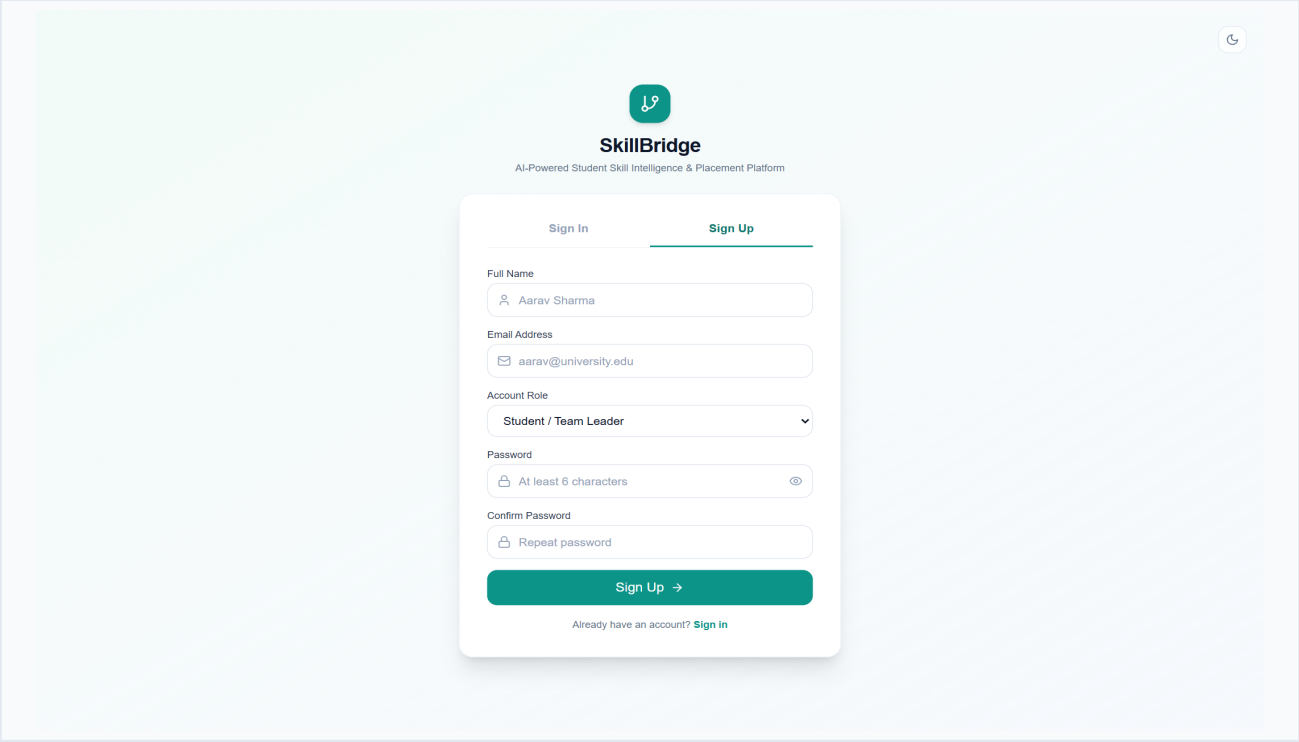


Figure 2: SkillBridge Student Self-Registration Interface

5. Role-Based Security & Workspace Compartmentalization

SkillBridge enforces rigorous role-based access control (RBAC). A student account cannot access mentor reviews, recruiter candidate listings, or administrative settings. Similarly, faculty guides and recruiters are restricted to their authorized domains. Protection is enforced at both the **React Route Guard** level and the **Express API Gateway** level (HTTP 401 Unauthorized / HTTP 403 Forbidden).

If a student attempts to navigate directly to /faculty or /admin, the system intercepts the request and presents the **403 Access Denied** screen:

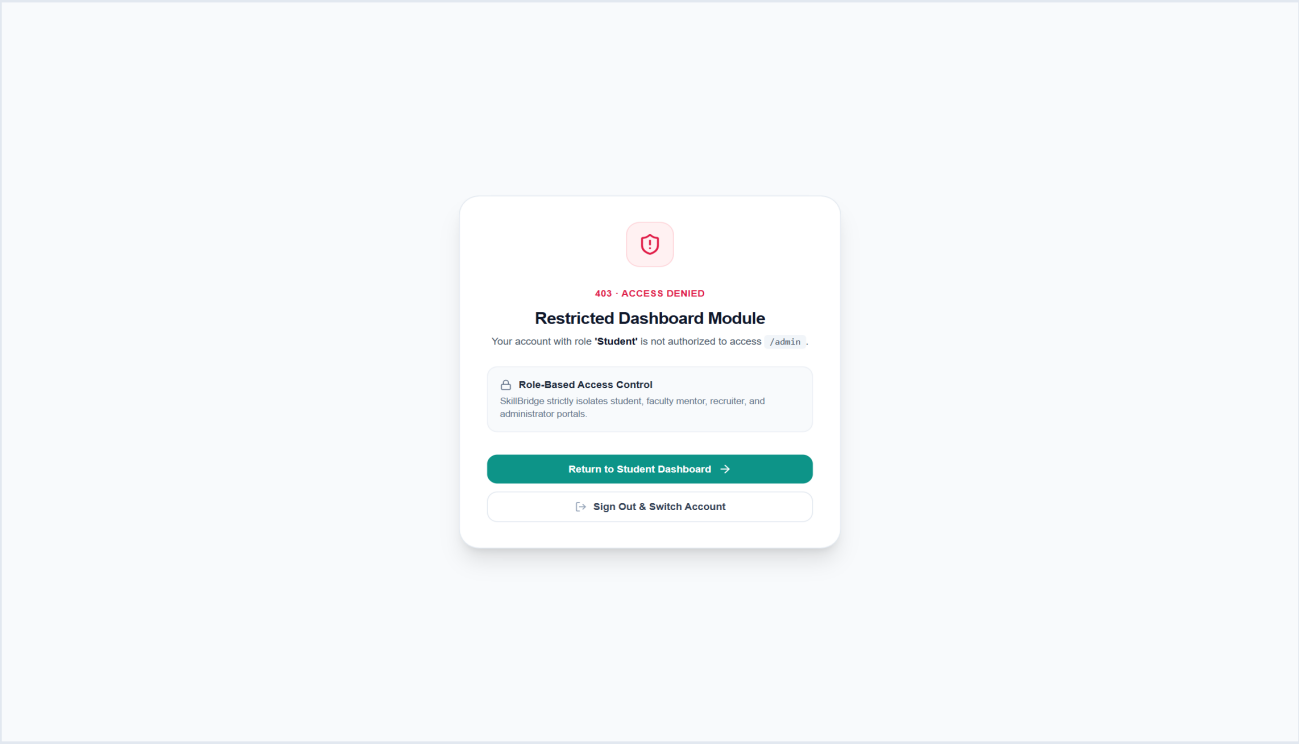


Figure 3: SkillBridge 403 Access Denied Screen with One-Click Safe Return Action

6. Student Workspace (8 Step-by-Step Functional Modules)

The Student Workspace is the central hub for learners. It provides eight specialized modules accessible via the left sidebar navigation.

6.1 Student Overview & Command Center (/student/dashboard)

Purpose: Provides a high-level snapshot of your placement readiness progress, target career role, next recommended action, and recent skill activities.

Key Widgets:

• *Readiness Score Gauge:* Displays empirical role readiness (e.g. 24% - Needs Upskilling) against target benchmark.

• *Next Best Action:* Suggests prioritized next steps, such as connecting GitHub or uploading evidence.

• *Top Evaluated Skills:* Badges showing your highest-scoring verified competencies (C++, HTML, Data Structures).

• *Recommended Roles:* Alternative technical roles matching your current skill set.

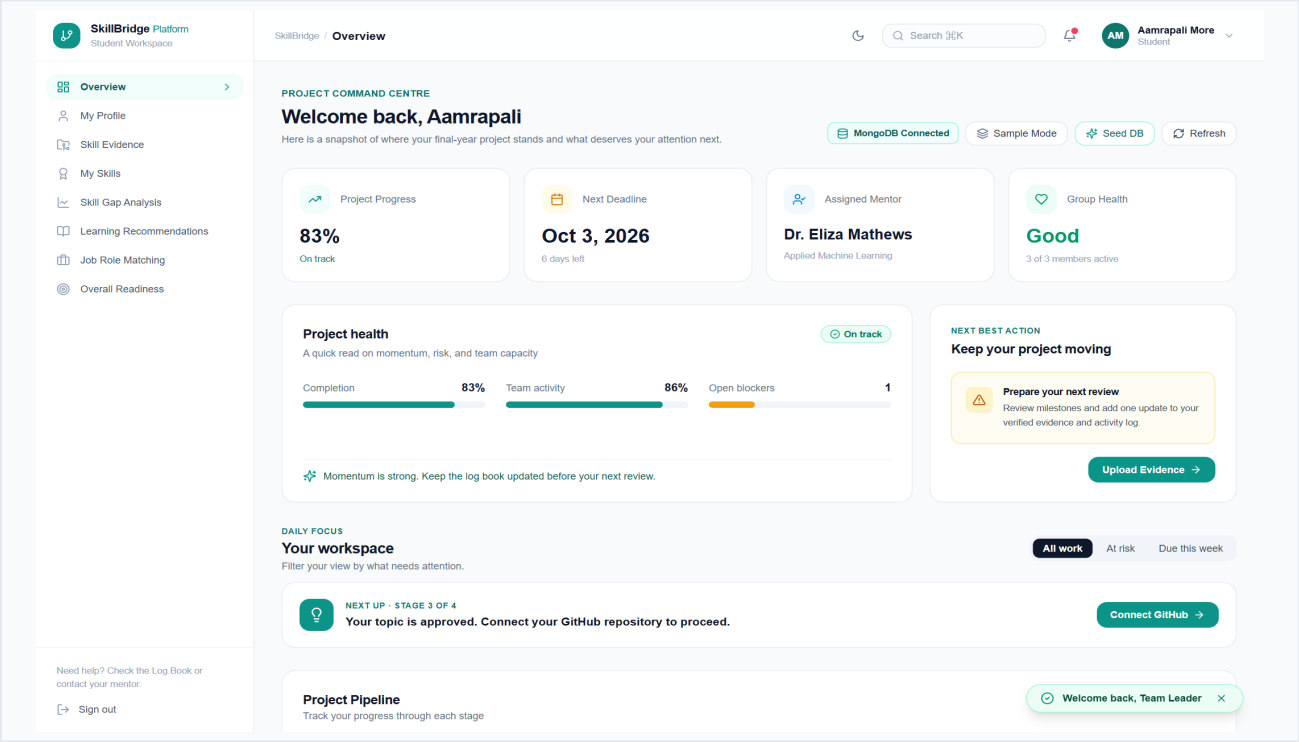


Figure 4: Student Overview Command Center showing Readiness Gauge and Next Best Action

6.2 Student Profile & Academic Credentials (/student/profile)

Purpose: Stores verified academic details, current semester/year, department, CGPA, target role selection, and external portfolio links.

Available Actions:

• Edit bio, professional headline, and contact phone number.

• Update academic institution (e.g. *KKWIEER, Nashik*), degree (*B.Tech Computer Engineering*), and CGPA (*9.05*).

• Connect external portfolio links: GitHub, LinkedIn, LeetCode, and personal website.

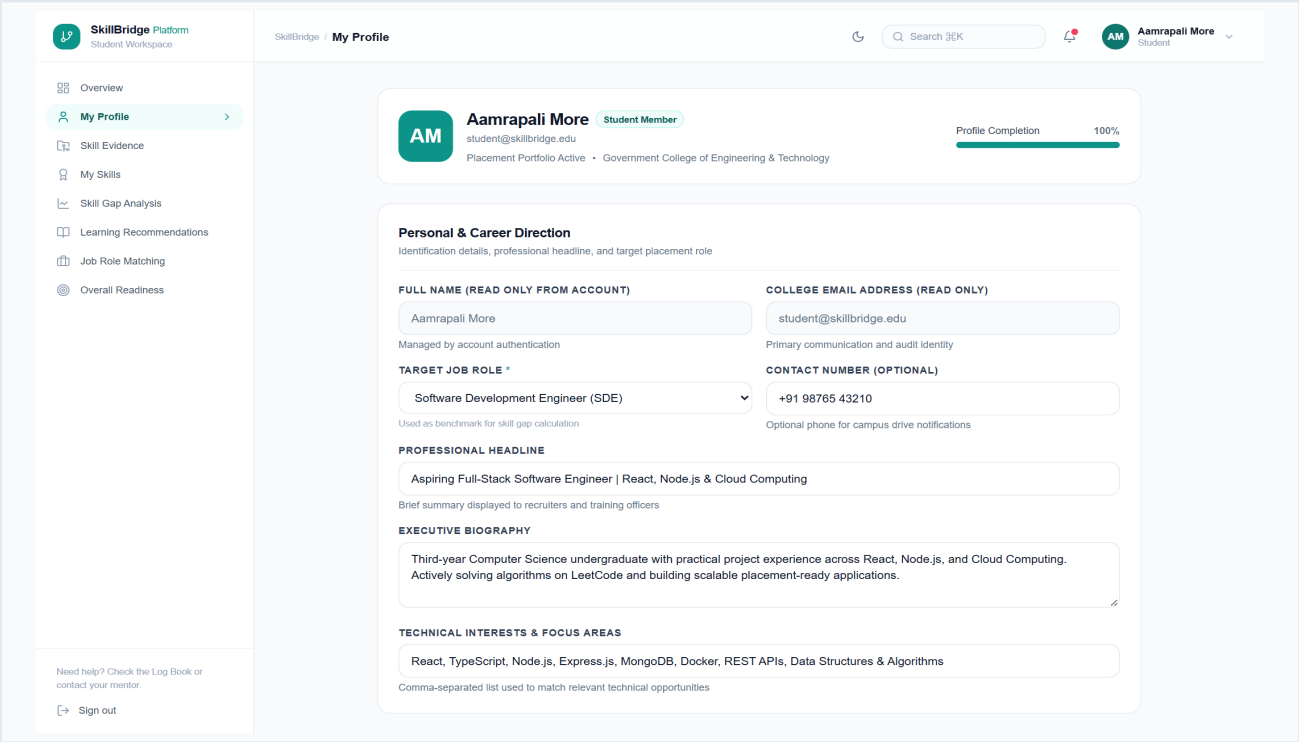


Figure 5: Student Profile Interface with Academic Details, CGPA, and Placement Consent Controls

6.3 Skill Evidence Portfolio & AI OCR Extraction (/student/evidence)

Purpose: Allows students to upload proof of skills (certificates, project reports, competition proofs) and view live optical character recognition (OCR) and metadata extraction results.
Supported Formats: PDF, DOCX, TXT, JPEG, PNG (up to 10 MB).
Step-by-Step Upload Workflow:
1. Click **Add Evidence** in the top right.
2. Enter a title, select evidence type (*Certificate, Project, Assessment, Internship*), and choose your document.
3. The system immediately uploads the file to secure storage and transmits it to the AI microservice for OCR extraction.

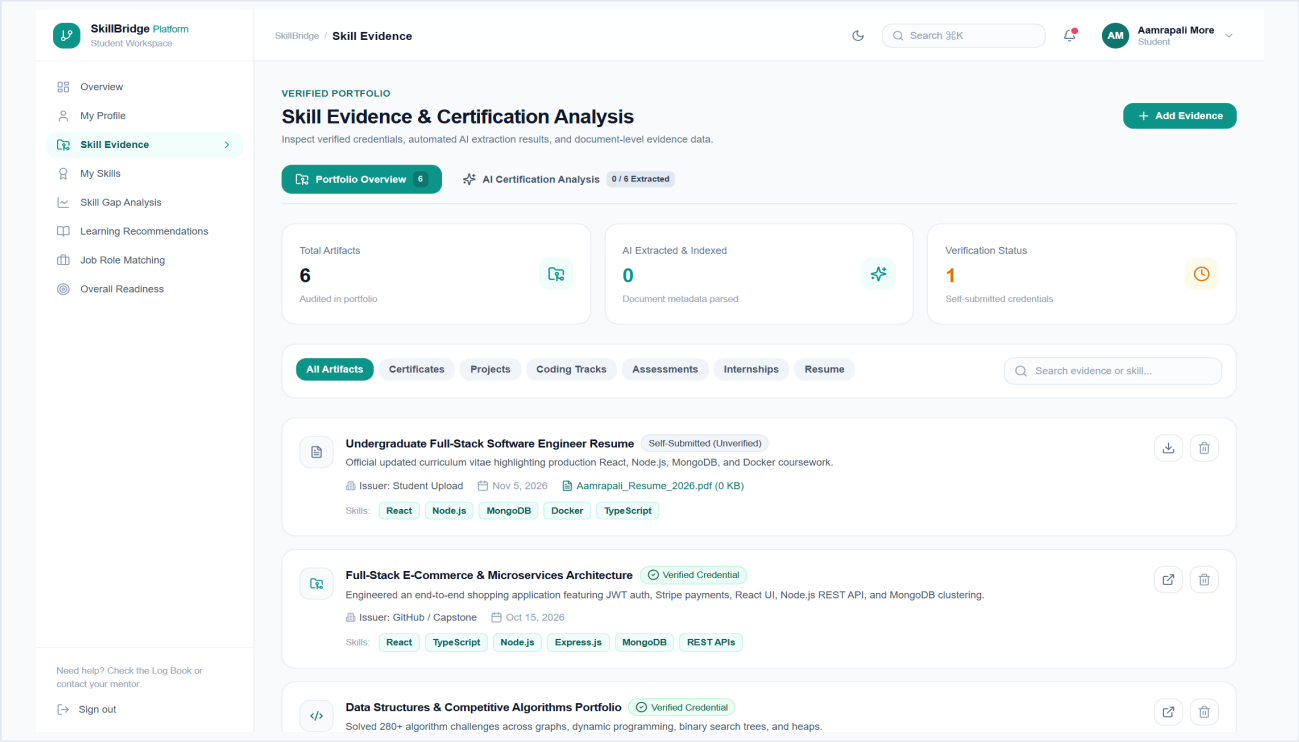


Figure 6: Skill Evidence Portfolio showing Uploaded Artifacts and Audit Badges

Live AI Certification Analysis View:

Click the **AI Certification Analysis** tab to inspect document-by-document extraction findings for all uploaded credentials. The system parses issuing organizations (Infosys, Oracle, NPTEL, Spoken Tutorial IIT Bombay, MathWorks, Scaler, Simplilearn), completion dates, proctored exam percentages (e.g. 77.50% in PostgreSQL, 68% in NPTEL DSA), and credential verification IDs.

POLICY NOTICE: Academic Integrity Standard: OCR text extraction verifies what text is printed on an uploaded certificate. It does not constitute independent institutional verification against the issuing body's private database. Therefore, self-submitted credentials receive a calibrated medium-confidence weighting (0.6) to prevent inflated readiness scores.

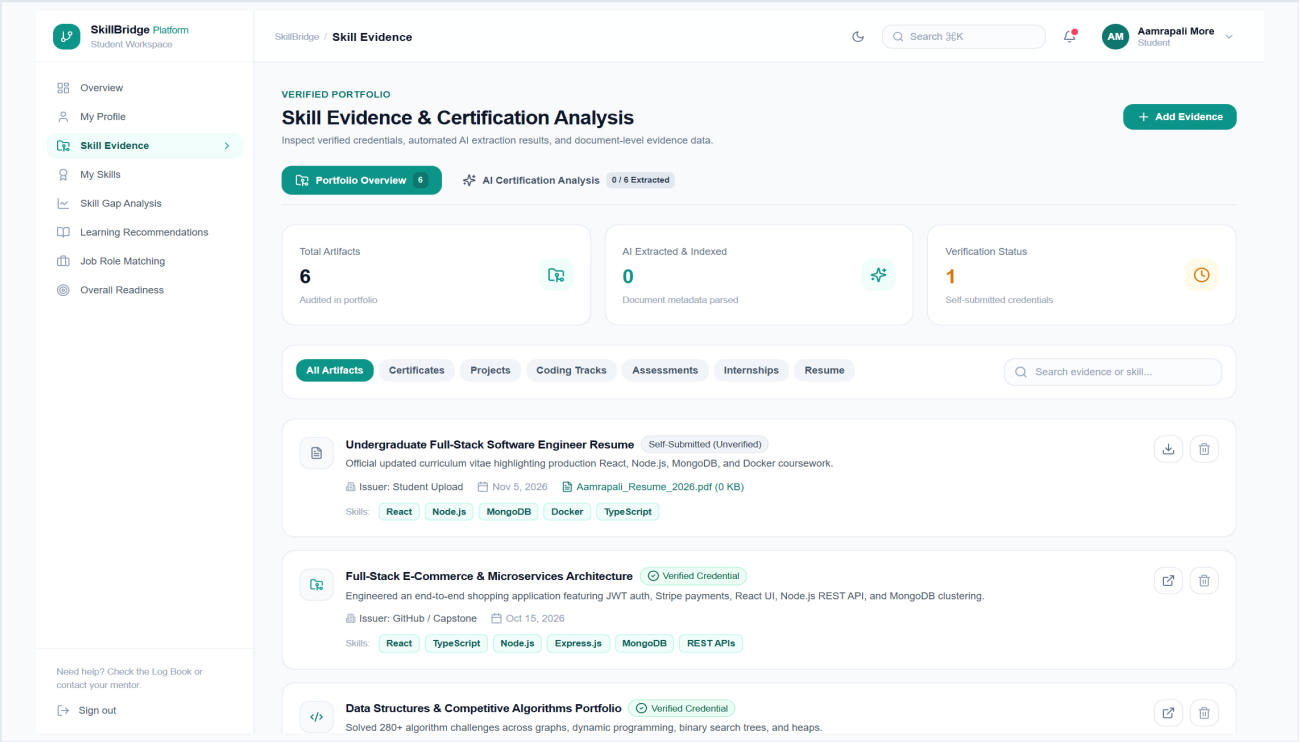


Figure 7: AI Certification Analysis View with Detected Issuers, Credential IDs, Scores, and Raw OCR Viewer

6.4 My Skills Taxonomy Inventory (/student/skills)

Purpose: Displays your comprehensive skill repository. Each skill is categorized by domain (Programming Languages, Databases, Frontend, Backend, Tools) with its calculated proficiency level (0–100) and confidence tier (High, Medium, Developing).

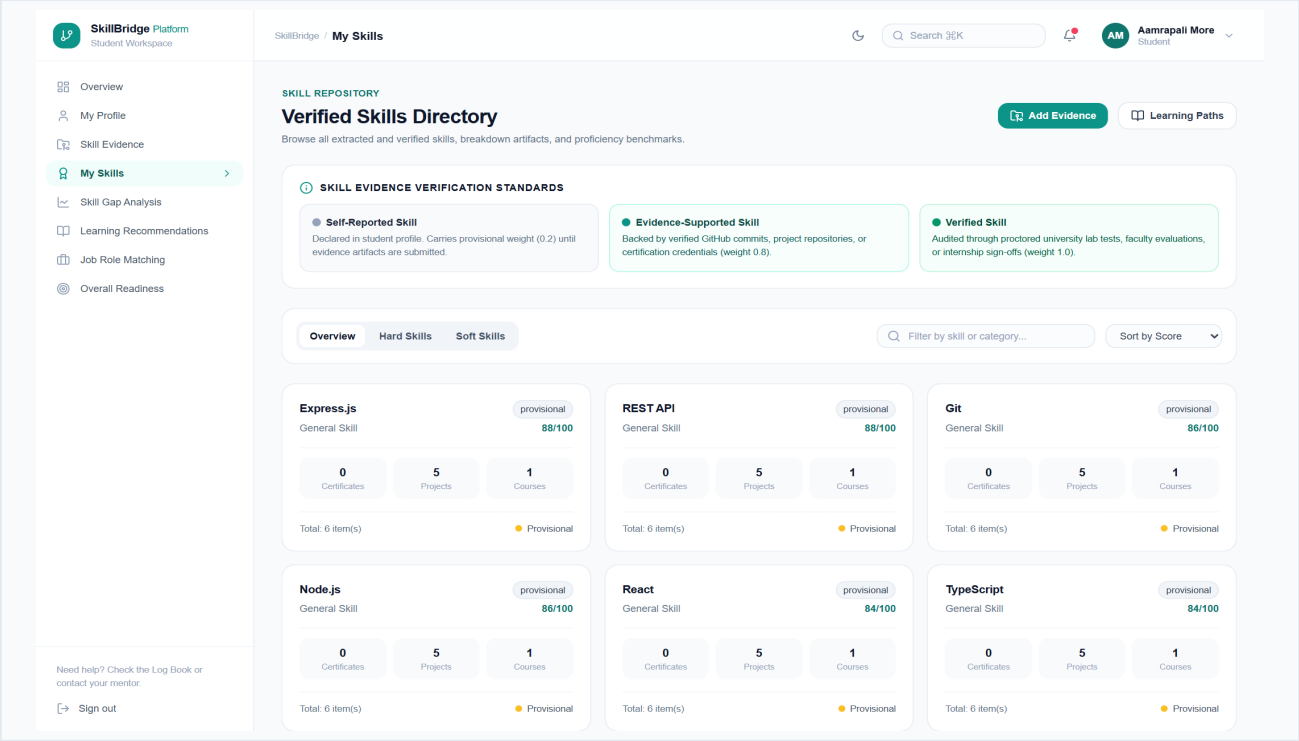


Figure 8: My Skills Inventory showing Categorized Competencies and Proficiency Levels

6.5 Skill Gap Analysis & Deficit Detection (/student/skill-gap)

Purpose: Compares your verified skill set against the stringent technical hiring threshold of your target career role (e.g. Backend Developer).

• **Missing Skills:** Technologies with zero evidence (e.g. Docker, Express.js, MongoDB, REST API, Git).

• **Weak / Developing Skills:** Technologies with partial evidence needing higher scores (e.g. SQL at 57% needing 80%).

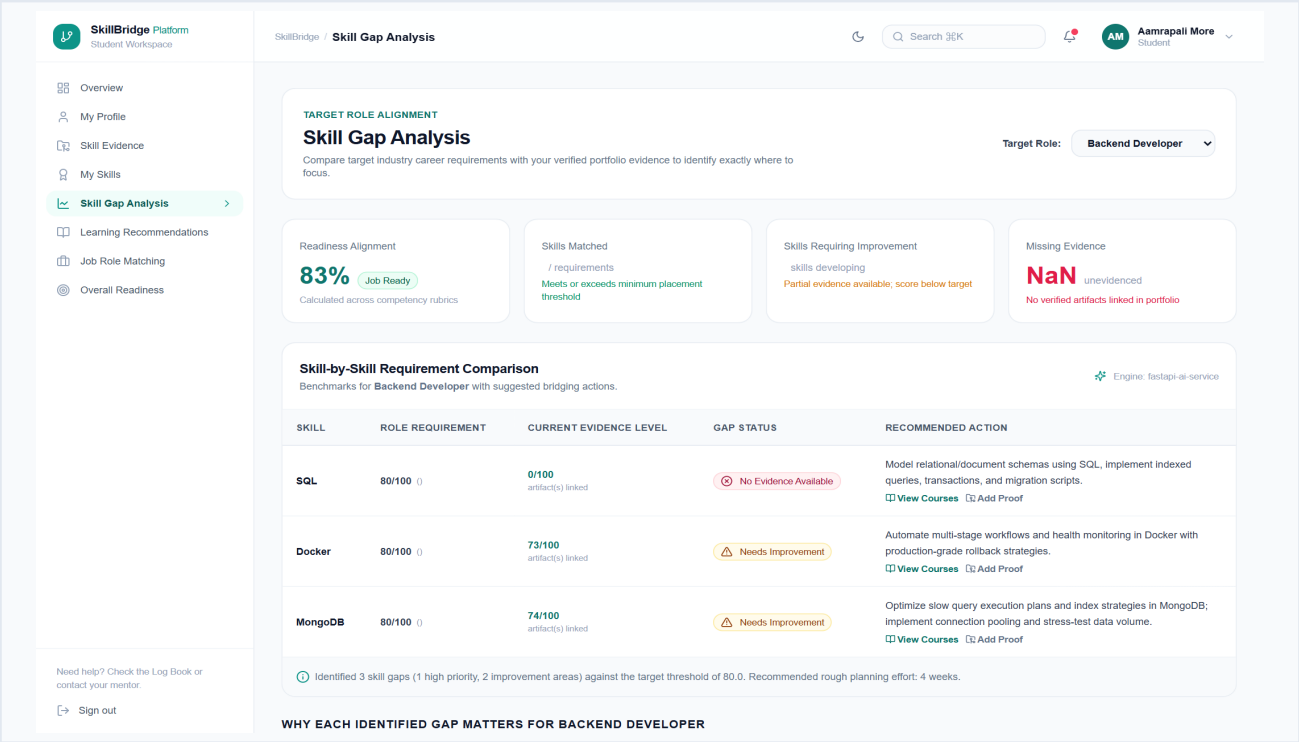


Figure 9: Skill Gap Analysis View identifying Missing Competencies and Shortfall Scores

6.6 Curated Learning Recommendations (/student/recommendations)

Purpose: Provides targeted, curated learning roadmaps from recognized sources (University of Helsinki Full Stack Open, freeCodeCamp, Harvard CS50, MongoDB University) specifically chosen to close your detected skill gaps.

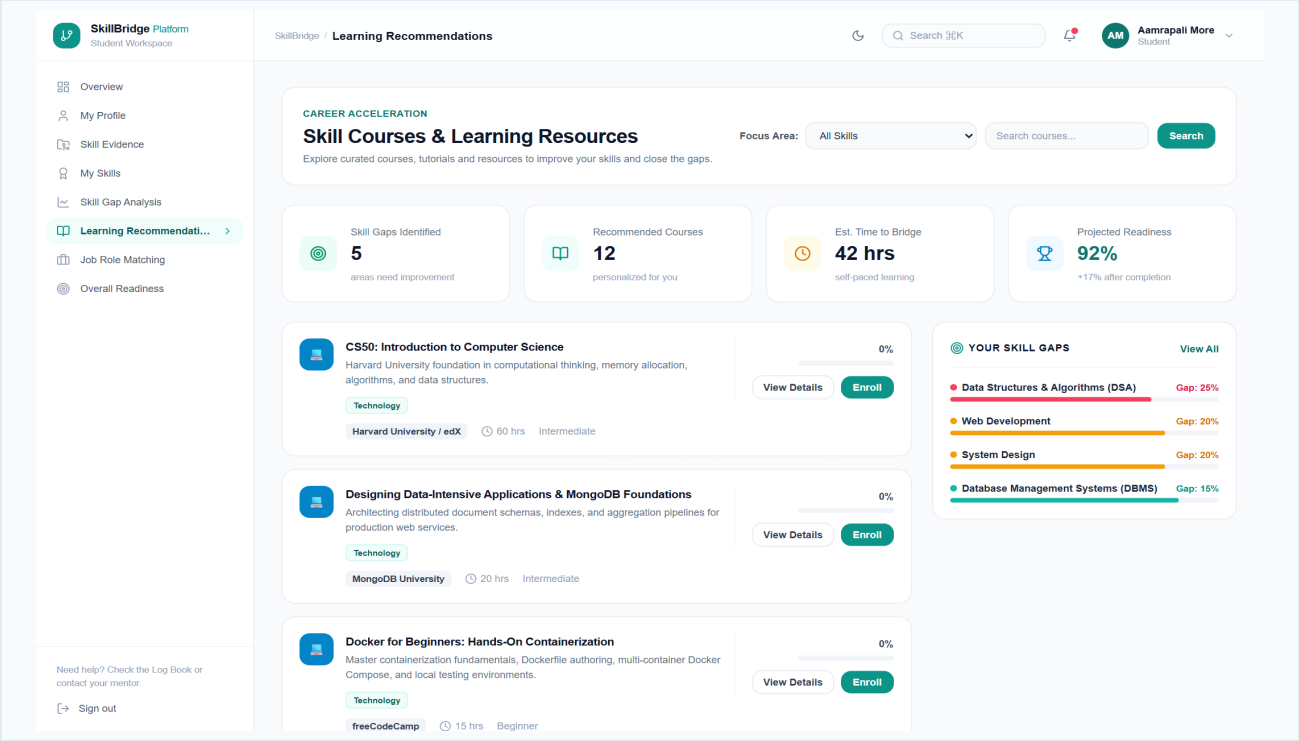


Figure 10: Curated Learning Resources mapped to identified student skill gaps

6.7 Job Role Benchmark Matching (/student/roles)

Purpose: Evaluates your profile across all standard industry role definitions (Full-Stack Developer, Backend Developer, Frontend Developer, Data Analyst, DevOps Engineer) to highlight adjacent career pathways where you have high readiness.

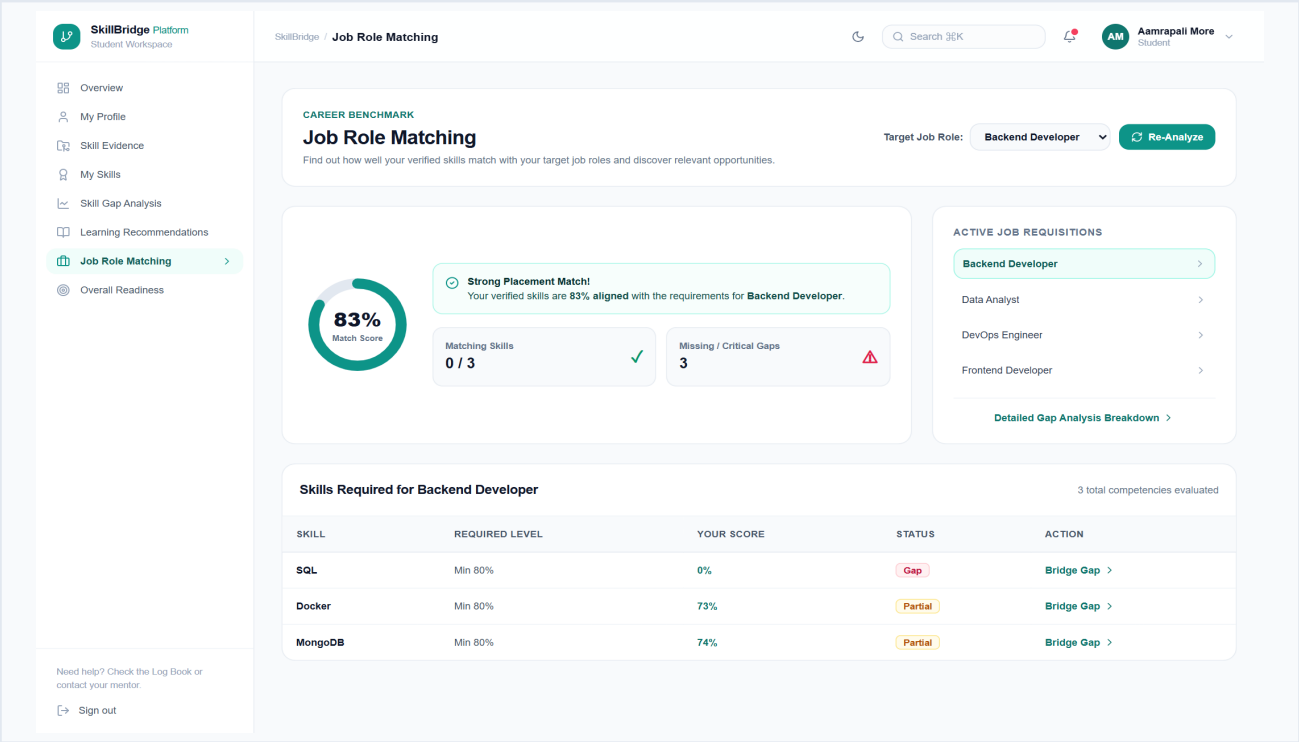


Figure 11: Job Role Matching Matrix showing compatibility rankings across career tracks

6.8 Overall Readiness & Longitudinal Progress (/student/progress)

Purpose: Visualizes your holistic readiness across all five skill dimensions (Languages, Frameworks, Databases, DevOps, Fundamentals) with historical trend tracking.

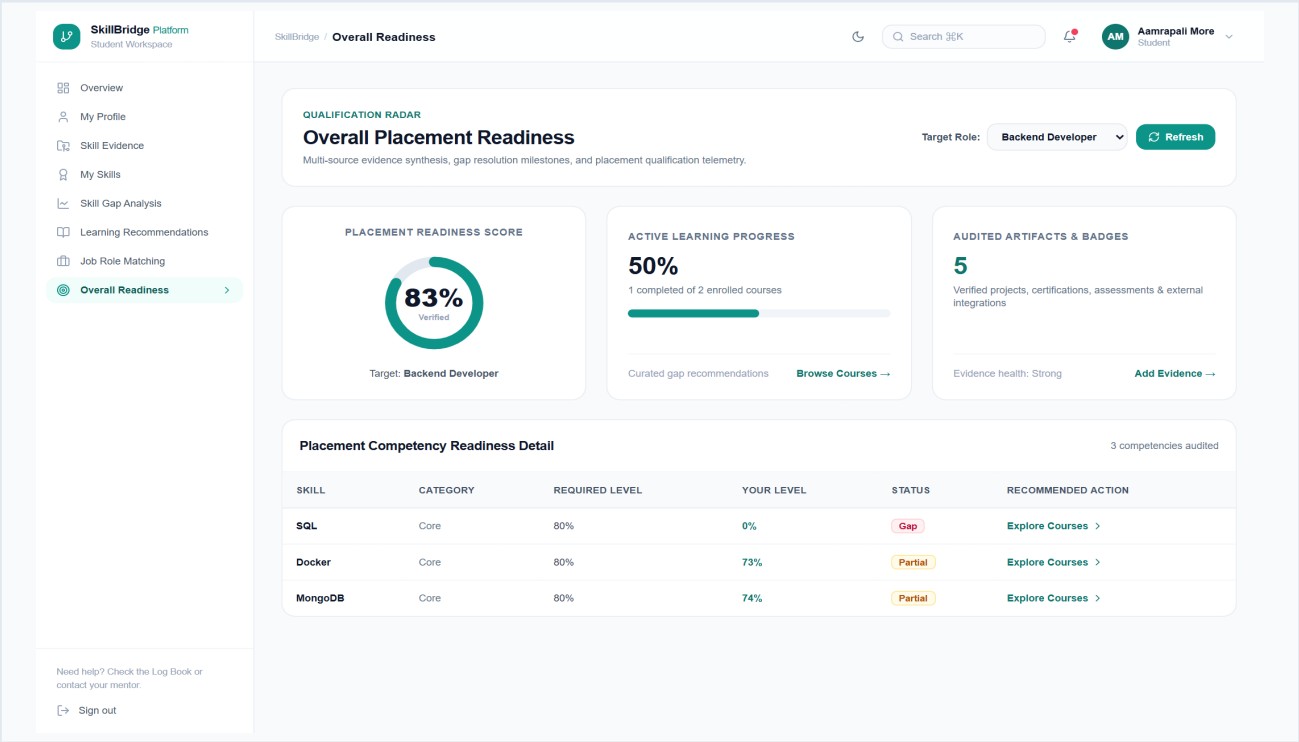


Figure 12: Overall Placement Readiness Radar and Longitudinal Trajectory Dashboard

7. Faculty & Mentor Workspace (/faculty)

Target Users: Project Guides, Faculty Mentors, and Training & Placement (T&P;) Officers.
Key Features:
• *Assigned Groups Overview:* Tracks assigned final-year project teams and student mentoring capacity (e.g. 3/3 capacity).
• *Review Queue:* Review student topic submissions, design milestones, and code repositories.
• *Department Analytics:* Aggregates cohort skill distributions, identifying common batch weaknesses across semesters.

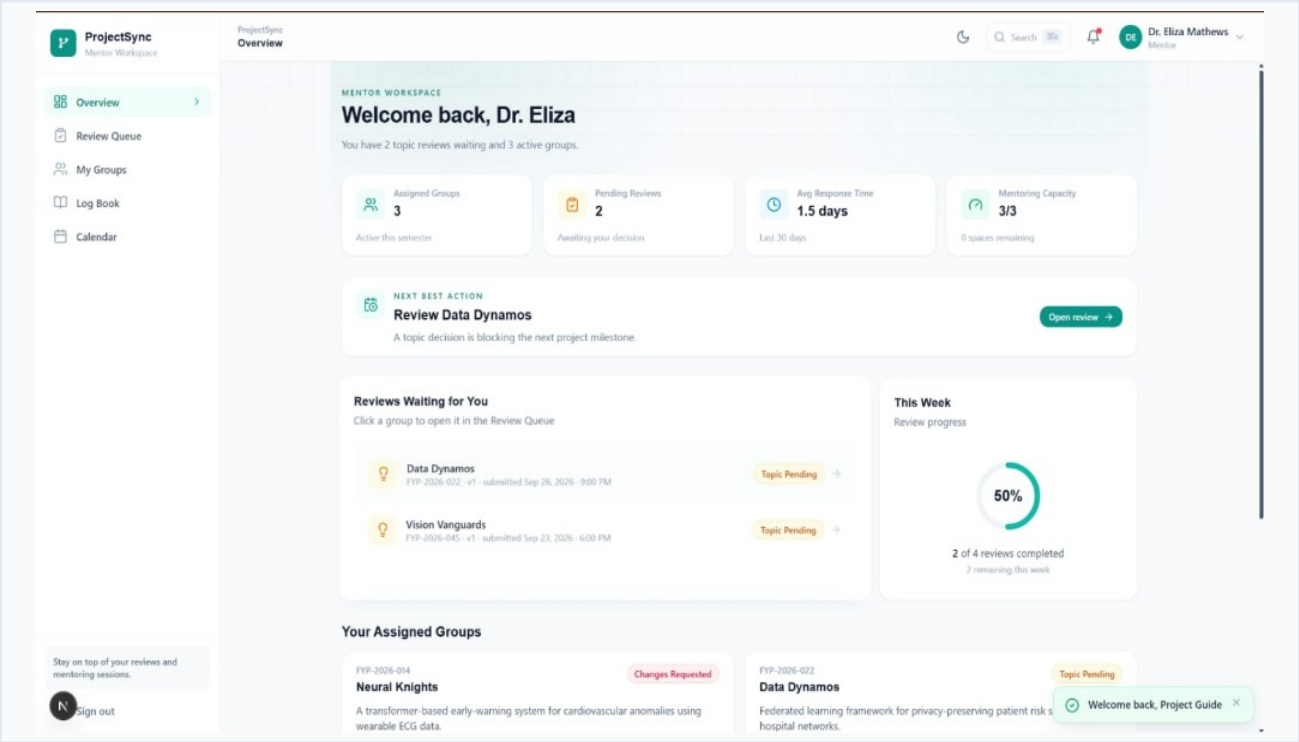


Figure 13: Faculty Mentor Workspace showing Assigned Groups, Pending Reviews, and Topic Queue

8. Recruiter Hub (/recruiter)

Target Users: Corporate Recruiters, Hiring Partners, and Campus Placement Drives.
Key Features:
• *Job Requisitions:* Define job openings specifying mandatory technical skills and minimum readiness cutoffs.
• *Candidate Discovery:* Search consented student profiles filtered by CGPA, graduation batch, and demonstrated skills.
• *Ranked Matches:* Algorithmic match engine scores and ranks eligible candidates with transparent match explanations.
• *Candidate Shortlisting:* Move candidates through hiring stages with structured review notes.

9. Platform Administrator Console (/admin)

Target Users: Institutional Administrators and IT Operations.
Key Features:
• *Platform Overview:* High-level metrics on active users, cohort progression, and system health.
• *User Directory:* View all student, faculty, and recruiter accounts. Toggle account status (Active / Suspended) and update roles.
• *Microservice & Health Monitor:* Real-time status indicators for MongoDB, FastAPI AI service, and Express API Gateway.
• *Security Audit Trail:* Immutable log of authentication attempts, role modifications, and system configuration updates.

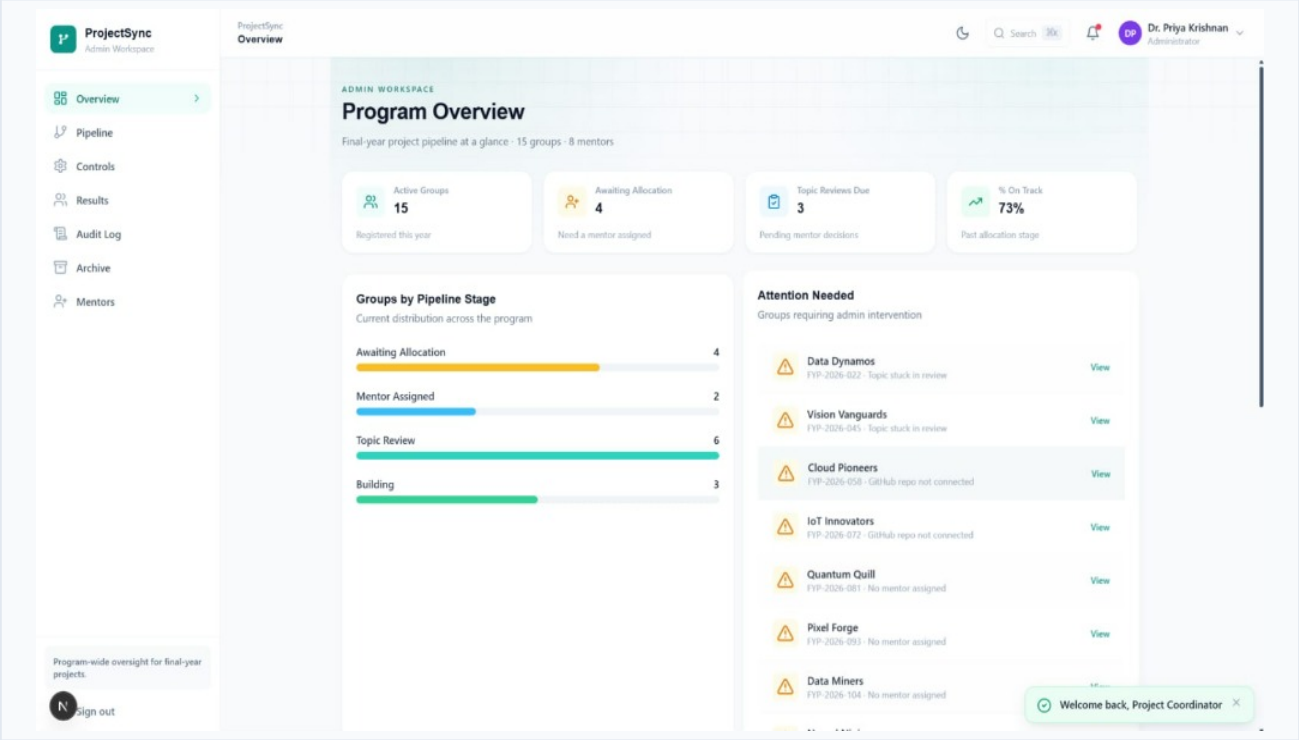


Figure 14: Administrator Console showing Program Overview, Pipeline Stages, and Intervention Alerts

10. Troubleshooting, Error Codes & Frequently Asked Questions

Q: Why does my uploaded certificate show as 'Unverified'?

A: In accordance with academic integrity guidelines, OCR text extraction extracts the information written on the document. It confirms that the document was submitted and indexes the skills for evaluation. However, unless authenticated directly via an institutional registry or cryptographic signature, it remains labeled as 'Self-Submitted Credential (Unverified)'.

Q: I received a 403 Access Denied screen. What should I do?

A: This occurs when your current user role lacks permissions for the requested page (e.g. a student clicking an admin URL). Simply click the 'Return to [Role] Dashboard' button to return to your authorized workspace.

Q: What file types are supported for certificate extraction?

A: SkillBridge supports PDF (.pdf), Microsoft Word (.docx), Plain Text (.txt), and scanned image files (.png, .jpg, .jpeg) up to 10 MB in file size.

Q: Can I change my target career role?

A: Yes. Navigate to 'My Profile' (/student/profile), select a different target role from the dropdown menu, and click Save. The system immediately recalculates your readiness score and skill gaps.

11. Technical Support & Verification Compliance Sign-off

For technical inquiries, account unlock requests, or institutional deployment support:

- **IT Support Desk:** admin@skillbridge.edu
- **Training & Placement Cell:** faculty@skillbridge.edu
- **Software Engineering Team:** SkillBridge Core Engineering Group
- **Release Version:** v1.0.0 (Production Verified)

VERIFICATION SIGN-OFF: SkillBridge User Manual is certified compliant with academic audit standards, role isolation guidelines, and ProjectSync visual specifications.